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The practice of cyber-threat intelligence in organizations: A socio-technical case study of a mature financial organization

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ABSTRACT

This socio-technical case study examines the operationalization of cyber-threat intelligence in a large regulated financial institution. Our findings reveal a fundamental inversion of established intelligence doctrine - rather than strategic requirements flowing downward from leadership to drive intelligence operations, our case organization generates requirements within technology operations silos, pushing intelligence outward and upward to operational, tactical, and strategic levels. This constitutes a reversal of intelligence norms established in military practice, revealing the unique challenges modern civilian organizations face in operationalizing CTI within complex business environments. Our study reveals a paradox where organizations invest heavily in CTI to address cyber-threats as a strategic priority yet simultaneously create barriers that prevent CTI from delivering strategic value. The root cause of this paradox is the positioning of the CTI function in the lowest level of the organizational structure, the profound knowledge gap between Business and IT Groups, and the prevalence of analytical products that lack strategic relevance or analytical rigor. We construct a stakeholder activity process model that contributes to theory by providing the first systematic empirical explanation of CTI practice in organizations and by applying intelligence doctrine to the cyber context. From a practice perspective, the model serves as a diagnostic tool for understanding bottom-up operationalization of CTI practice. It enables organizations to map intelligence flows arising from key stakeholders, processes, and technology systems, and to redesign these flows to improve strategic coherence and relevance.

1. Introduction

The cyber-threat landscape has rapidly evolved in recent years with the introduction of organized, sophisticated, and well-resourced threat actors in the form of organized crime syndicates and paramilitary cyber entities backed by nation-states. Cyber defenses have not evolved at the same pace. Even the most well-resourced commercial organizations, typically in the financial sector, burdened by complex law, regulation, and 'best practice' industry frameworks, lack the capability to defend against such Advanced Persistent Threats (APTs) (Ahmad et al., 2019; Chen, 2014; Gilbert, 2025).

Cyber-threat intelligence (CTI) has a critical role to play by assisting organizations to understand the threat actors (e.g., their objectives, behavior, and tactics) to better direct their cyber defense practices. CTI is a critical decision-support function that promises to transform cyber defense from reactive and undirected responses to proactive,

anticipatory and directed capabilities. Hence CTI enables organizations to anticipate the intent and capability of threat actors, thereby facilitating the development of defensive strategies to counter expected attack patterns. In pursuit of this transformation, many organizations have reportedly invested in threat intelligence platforms and dedicated CTI teams (Brown and Sfakianakis, 2025).

Yet despite cybersecurity being consistently ranked among top enterprise risks and the substantial organizational investment driven by strategic risk concerns, there is very little empirical evidence about how organizations actually implement and operationalize intelligence-led approaches in complex socio-technical environments. To understand how CTI can be operationalized in organizations, we draw on intelligence doctrine situated in the military context as theoretical underpinning, as this represents the most established approach to intelligence-led decision-making. Using the intelligence cycle as our theoretical lens we examine a mature CTI practice in a leading Australian financial

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organization. The empirical evidence we collect is framed as a stakeholder-activity process model that explains how CTI is practiced across Leadership, Business, and Technology Groups.

Our review of professional and academic literature found few studies that engage in a rigorous examination of the practice of CTI in organizations. While extensive literature addresses new CTI technologies and information sharing mechanisms (Ainslie et al., 2023; Tounsi and Rais, 2018; Wagner et al., 2019), the actual practice of CTI in real-world organizations remains largely unexplored. In fact, aside from a clinical study of CTI-as-a-service (Kotsias et al., 2022), there are few, if any, studies of practice centered on CTI, as pointed out in a comprehensive review of CTI research (Ainslie et al., 2023).

We therefore pose the question: “How do organizations practice CTI?”

To answer this question, we introduce the intelligence doctrine from the military context to understand how cyber threat intelligence practice has previously been designed to support strategic decision-making. We then review the academic and practitioner literature on CTI and reflect on the lack of practice-based research. Subsequently, we present our research approach focusing particularly on why we adopted a single-case study approach, the intelligence process as kernel theory, and a stakeholder-activity process model as the primary vehicle to explain CTI practices in our case organization. The findings from our case study are then presented, along with our constructed stakeholder-activity process model. We then engage in a deep discussion of the root causes behind the findings. The paper concludes with contributions to both theory and practice, as well as limitations and future work.

2. Background

2.1. The role and function of intelligence in organizations

In layperson terms, the role of the intelligence function in an organization is to support decision-making by providing the right information to the right people at the right time. Intelligence, in its doctrinal form and relevant to the context of security threats, represents “*information and knowledge about an adversary obtained through observation, investigation, analysis, or understanding*” that enables decision advantage (JCS, 2013). Critically, intelligence is not merely information, but rather information that has undergone rigorous analytical processes to support specific decision-making needs (Lowenthal, 2020).

The **Intelligence Cycle** establishes a requirements-driven process where strategic priorities cascade through organizational structures (MOD, 2023). The cycle begins with Direction, where leadership establishes intelligence requirements based on strategic objectives. These requirements drive Collection activities, focusing resources on gathering specific information needed for decisions. Processing transforms raw information into intelligence through structured analytical techniques and merging new intelligence with previously processed and stored extant material. Finally, Dissemination delivers tailored intelligence products to decision-makers who initiated the requirements (Glass, 1948; Kent, 1964).

In the military context, top-down flow is not arbitrary but fundamental to intelligence effectiveness. As Herman (1996/1996) established in his seminal analysis of intelligence power, the connection between intelligence and strategic decision-making determines whether intelligence provides value or merely generates noise. When intelligence operates bottom-up (a possibility Herman warned against but considered unlikely in military contexts), it loses strategic relevance and becomes “an expensive encyclopedia” rather than a decision-support capability.

The importance of analytical rigor in distinguishing intelligence from information cannot be overstated. Heuer (1999) documents how structured analytical techniques protect against cognitive biases and analytical failures. These techniques, from Analysis of Competing Hypotheses to Key Assumptions Checks, require extensive training and disciplined application. Military intelligence officers typically undergo

months or years of training before conducting independent analysis, establishing analytical standards that become institutional knowledge (Marrin, 2011).

Having established intelligence flows top-down in military organizational contexts, it remains to be seen what flow of CTI would be effective in the modern and complex business environment. This is a key question that our study aims to address.

Historical intelligence doctrine dating to 1948 (Glass, 1948) through contemporary publications emphasizes that intelligence “requires constant review and updating” (Army, 2018, p. 12) achieved through feedback from decision-makers to collectors, from consumers to producers, and among peer analysts. The Intelligence Cycle phases do not operate in isolation but are connected through continuous feedback mechanisms that transform the intelligence cycle from a unidirectional production process into an iterative learning system (Army, 2018; MOD, 2023).

Military doctrine conceptualizes this feedback in multiple forms: “evaluative feedback” where intelligence consumers assess the relevance, accuracy, and timeliness of products and communicate deficiencies back to producers; “requirements refinement feedback” where initial intelligence answers generate new or adjusted questions that redirect collection and analysis; “quality feedback” where collectors and analysts validate information sources and analytical approaches based on product utility; and “peer feedback” where horizontal communication among analysts and between organizational units improves shared understanding and identifies gaps (MOD, 2023, pp. 39–40); (Army, 2018, pp. 12–13); (Chismon and Ruks, 2015).

This feedback is represented variously across doctrinal traditions as ‘evaluation’ (US doctrine), ‘requirements refinement’ (NATO doctrine, see fig 3.1 JDP 2–00, p. 38), or ‘continuous communication and review’ (UK/Australian doctrine), but all share the fundamental principle that effective intelligence cannot be produced without iterative dialogue between producers and consumers. Doctrinal diagrams consistently depict these feedback mechanisms as surrounding or connecting all phases rather than as a discrete sequential step, emphasizing their continuous and pervasive nature throughout intelligence operations.

2.2. The utility challenge - CTI emergence and evolution

CTI research is generally conducted from two perspectives – the technology perspective (Chismon and Ruks, 2015), and the information-sharing perspective (Tounsi and Rais, 2018). A third perspective is the organizational perspective that has only recently emerged in the literature (Ahmad et al., 2021; Ettinger et al., 2019; Kotsias et al., 2022; Ramsdale et al., 2020), with a comprehensive examination revealing that ‘CTI literature is dominated by the technology discourse (Ainslie et al., 2023, p. 2)’.

2.2.1. The technology perspective – intelligence as data processing

The overwhelming majority of CTI literature approaches intelligence as a technological capability. Studies focus on threat intelligence platforms (TIPs), automated indicator processing, and technical standards for threat data sharing (Dandurand and Serrano, 2013; Sauerwein et al., 2017; Wagner et al., 2019). This perspective frames intelligence primarily as a data problem: collecting more indicators, processing them faster, and distributing them more efficiently.

Recent developments have intensified this technological focus. Moraliyage et al. (2022) propose AI-enhanced CTI processing pipelines that promise to automate the intelligence cycle. While acknowledging that “AI can recognize patterns, human analysts understand the broader context”, the emphasis remains on technological solutions to what are fundamentally analytical challenges. The 2024 SANS CTI Survey reports that 78% of organizations prioritize technology investments over analytical training, suggesting this perspective dominates practice (Brown and Sfakianakis, 2024).

Framing CTI as a technology problem with a technology solution

fundamentally misunderstands intelligence. As Chismon and Ruks (2015) observe in their classification of strategic, operational, tactical, and technical intelligence, the focus on technical indicators represents only the lowest level of intelligence, yet it receives the most attention in literature and practice. The assumption that sophisticated technology equals intelligence capability (what we term the ‘technology sophistication trap’) emerges directly from this perspective.

2.2.2. The information sharing perspective: intelligence as collective knowledge

A second stream of literature frames CTI primarily as an information-sharing challenge. Research emphasizes standards like STIX and TAXII, sharing platforms offered through ISACs,¹ and trust mechanisms for cross-organizational collaboration (Barnum, 2012; Burger et al., 2014; Wagner et al., 2016). This perspective assumes that pooling threat information across organizations creates intelligence through collective awareness.

However, recent research reveals fundamental challenges with this assumption. The CTI Networking Report (Chi, 2024) found that despite extensive sharing mechanisms, “*numerous organizational challenges came to light, like issues of legal liability, a lack of formalized processes, and gaps in measuring effectiveness.*” More critically, Abu et al. (2018) demonstrate that shared threat data often lacks the context necessary for strategic decision-making, reducing intelligence to tactical indicator matching.

The information-sharing perspective conflates information exchange with intelligence production. It is important to note that while sharing can enhance collection, it cannot replace the analytical processes that transform information into intelligence.

2.2.3. The organizational practice perspective - the missing link

The least developed but most critical perspective for this study examines how organizations actually implement and use CTI. This gap in empirical research is striking given CTI’s practical objectives. While Kotsias et al. (2022) provide valuable insights into CTI adoption challenges, documenting the transformation from “*compliance-driven and reactive logic towards a proactive logic driven by CTI*” their action research approach focuses on describing a particular challenge faced by CTI rather than systematic examination of an organization’s holistic practices.

Ramsdale et al. (2020) surveyed CTI maturity across organizations, finding that most struggle with “*translating intelligence into actionable insights*” and “*demonstrating value to leadership*”. Yet the survey could not explain why these struggles persist despite technological sophistication and process implementation. Similarly, Ettinger et al. (2019), in a study of Cyber Intelligence tradecraft, identified capability gaps but lacked empirical data on actual practice within organizations.

This lack of empirical research on CTI practice is problematic given what Evans (2009) identified as the “*failure of imagination*” in intelligence, i.e., the inability to recognize when established frameworks no longer match operational reality. Without empirical examination, we cannot know whether CTI practice aligns with intelligence doctrine or has evolved into something fundamentally different.

2.2.4. Alternative CTI cycle conceptualizations

This study reviews how military intelligence doctrine has been adapted into civilian cyber-threat intelligence (CTI) practice, noting multiple ‘lifecycle’ models emphasizing different facets of the intelligence process. Tounsi and Rais (2018) articulate a six-phase lifecycle (Planning and Direction, Data Collection, Data Processing, Data Analysis, Intelligence Production and Dissemination, and Feedback), framing CTI as fundamentally data-centric and reflecting a practitioner focus on automated threat processing. Chismon and Ruks (2015), influenced by UK government practice, propose a five-phase model (Requirements,

Collection, Analysis, Production/Dissemination, Evaluation) and a “modified threat intelligence functional flow” that separates management functions (Requirements, Resource Management) from execution (Collection, Analysis), with Evaluation linking them. This distinction highlights organizational complexity in operationalizing intelligence.

Other authors (Bauer et al., 2020; Chantzios et al., 2019; Lee, 2020; Sauerwein et al., 2021) advance lifecycles emphasizing platform capabilities and technology implementation, while recent work stresses information sharing and platform enablement (Ainslie et al., 2023). Against this background, we justify using the classic four-phase military cycle (Direction, Collection, Processing, Dissemination) as our analytical framework for four reasons: it provides the analytical foundation from which most CTI variants diverge, enabling systematic comparison; its simplicity clarifies analytical boundaries; it aligns with our practice-focused aim to study decision-support rather than platform evaluation; and it provides a stable reference to reveal systematic divergence between military doctrine and civilian practice.

The literature’s multiple cycle formulations reflect attempts to reconcile doctrine with technological and organizational realities. Crucially, however, we argue that differences in phase count or technological emphasis are secondary to the pervasive absence of connective mechanisms, strategic direction, and feedback. Critically, the four-phase doctrine’s embedded feedback mechanisms that connect Direction to Collection, Collection to Processing, and Processing to Dissemination provide the analytical framework enabling us to identify where feedback connections should exist at FINORG (see Section 3.2). Our empirical examination (Section 5.2.2) demonstrates these doctrinal connections are systematically absent in practice. This feedback breakdown emerged as an empirical finding from applying military intelligence doctrine to civilian organizational practice, not as a predicted dysfunction. Whether feedback appears as an explicit fifth phase (Tounsi and Rais, 2018) or as cross-cutting connections (our four-phase model), the central finding remains in that feedback mechanisms enabling organizational learning are systematically broken in civilian CTI implementations.

2.2.5. The critical gap - no empirical examination of practice

Despite the substantial investment made by many organizations to develop their CTI function, few empirical studies exist that systematically examine how organizations actually practice CTI in complex business environments or what factors enable effective operationalization. Ainslie, et al. (2023) explicitly identify this gap in their research agenda for CTI, calling for empirical studies that examine “*how cyber-threat intelligence can be operationalized in organizations*”. They note that without understanding practice, we cannot determine if CTI delivers its promised value or merely creates an “*illusion of intelligence*” that may leave organizations more vulnerable than acknowledged ignorance.

This gap becomes critical when considering the stakes involved. Financial institutions face sophisticated threats from nation-states and organized crime groups with resources exceeding defensive capabilities (CrowdStrike, 2023). If these organizations operate with false confidence in compromised intelligence capabilities, believing they have intelligence-led security while operating blind, the consequences could be catastrophic, not just for individual institutions but for global financial stability.

Our research addresses this critical gap through the first systematic empirical examination of CTI practice. By investigating how a mature financial institution practices CTI across stakeholder groups and intelligence cycle phases, we reveal patterns that explain how organizations implement CTI programs in the modern business environment.

3. Research approach

3.1. Research design - the revelatory case study

Our investigation employed a single-case study design, selected for its capacity to provide deep, contextual understanding of real-world

¹ ISAC – Information Sharing Analysis Centers ([link](#))

practice within a complex organizational setting (Yin, 2018). The case study approach proves particularly appropriate for examining phenomena where the boundaries between context and practice are blurred, a critical consideration given CTI's embeddedness within organizational structures, cultures, and technologies.

We characterize this as a revelatory case study (Yin, 2018) for two main reasons. First, our research provides unprecedented access to CTI operations typically shielded from external scrutiny. Second, we explore a phenomenon (the actual implementation of CTI) that has not been studied before at this level of depth, despite extensive theoretical discussion.

The selection of a single in-depth case for this analysis was deliberate. Given the lack of prior empirical research on CTI practice, we required sufficient depth to understand not just what organizations do but why patterns emerge and how different elements interact. As Siggelkow (2007) argues, a single case can be powerful for demonstrating the existence of surprising phenomena that runs counter to what theory suggests should exist (in this case the theory is espoused in intelligence doctrine).

Our case selection follows Siggelkow's (2007) principle that single cases can be "rich" and "revelatory" when they provide unprecedented access to previously hidden phenomena. The term "revelatory" in case study research (Yin, 2018) designates situations where researchers gain access to organizational contexts previously closed to systematic inquiry. FINORG represents a revelatory case for two reasons: (1) unprecedented access to cross-functional stakeholders spanning leadership, business operations, and technology groups - rarely available for academic research in financial institutions, and (2) complete intelligence cycle coverage encompassing all phases of intelligence doctrine implementation rather than isolated components. The revelatory designation speaks to access depth and phenomenon novelty, not to statistical representativeness, a distinction fundamental to interpretive case research (Walsham, 1995). The case's theoretical generalizability, addressed in Section 3.2.1, derives from structural forces that shape CTI positioning across similar institutions rather than statistical sampling logic.

3.2. Case selection - finorg as exemplar

FINORG is an Australian multinational financial institution that was selected through purposeful sampling to represent a mature practitioner of CTI. FINORG spans multiple continents with substantial assets, making it a prime target for sophisticated threat actors, including nation-states and organized crime. This threat exposure has driven significant investment in cybersecurity capabilities, including a dedicated CTI program that has been operating for over five years. Of significant import is the unique access provided for the conduct of this study, an aspect that is remarkable and rare given the sensitive nature of FINORG operations. FINORG's generalizability derives from structural positioning common across the financial services sector rather than organization-specific attributes. Section 3.2.1 explains how institutional forces shape similar CTI configurations across large regulated financial institutions.

3.2.1. Institutional context and generalizability

The generalizability of single-case findings to the broader financial services sector is strengthened by the institutional forces that create structural similarities across large financial institutions. Institutional isomorphism, the process through which organizations in the same field become increasingly similar, operates through three mechanisms in this industry sector (DiMaggio and Powell, 1983).

FINORG's generalizability derives from its structural configuration rather than organizational identity. The organization positions CTI

within its Technology Group, reflecting an industry-standard model shaped by three isomorphic pressures. (1) Coercive isomorphism, from regulatory requirements directly shaped FINORG's CTI structure: [APRA Prudential Standard CPS 234 Information Security \(2019\)](#)^{2, 3} mandates information security capabilities positioned within operational risk frameworks, driving FINORG's placement of CTI as a security operations function rather than strategic capability. [Basel Committee on Banking Supervision - Principles for Operational Resilience \(2021\)](#) reinforces this positioning by framing cyber threat intelligence as technical risk management. (2) Normative isomorphism, through professional networks influenced FINORG's staffing and operational approach: the CISO and CTI team members hold common industry certifications (CISSP, CISM), participate in FS-ISAC⁴ intelligence sharing, and benchmark against peer financial institutions, including conformance with U.S. federal laws such as Sarbanes-Oxley (SOX)⁵, creating convergent perspectives on what CTI should look like and where it should sit organizationally. (3) Mimetic isomorphism, shaped FINORG's five-year program maturation: facing uncertainty about effective CTI practice, FINORG adopted structural patterns from peer organizations perceived as successful including a dedicated Threat Intelligence Platform, specialized team, Technology Group positioning, therein diffusing similar organizational arrangements across the sector (Haveman, 1993). Our findings emerge from this structurally determined configuration: CTI embedded in Technology, separated from Business by organizational boundaries, reporting through technical rather than strategic chains. Because these structural arrangements result from sector-wide isomorphic pressures rather than FINORG-specific choices, the dysfunctions we observe are likely products of common configurations across similar institutions. These isomorphic pressures operate most strongly in well-resourced, compliance-focused institutions. That dysfunction emerges at FINORG despite these favorable conditions suggests structural forces overwhelm organizational capability.

These isomorphic pressures create not just structural similarity but functional constraints that generate predictable dysfunctions. Regulatory isomorphism (APRA CPS 234, Basel Committee) mandates CTI positioning within operational risk frameworks, forcing Technology Group placement that prevents strategic access. This creates the requirements vacuum, leading to bottom-up direction. Professional isomorphism (CISSP, CISM, FS-ISAC) establishes a shared technical vocabulary that creates Business-Technology semantic boundaries, blocking cross-functional communication and generating parallel shadow intelligence systems. Mimetic isomorphism (peer benchmarking) diffuses 'best practice' technology platforms that automate and institutionalize bottom-up patterns, creating measurement illusions where activity metrics mask strategic irrelevance. The causal path reflects: regulatory pressure to operational positioning to strategic exclusion to requirements vacuum to technology self-direction to bottom-up inversion.

Independent empirical observations corroborate this pattern beyond FINORG. Kotsias et al. (2022) documented similar bottom-up intelligence dynamics at another Australian financial institution, where "CTI requirements are predominantly set by IT operations staff rather than business stakeholders," demonstrating technology-driven requirements generation consistent with our findings. Brown & Lee's (2021) SANS industry survey of financial services organizations found the same bottom-up requirements pattern across surveyed institutions, suggesting the phenomenon extends beyond individual cases. Ainslie et al.'s (2023) comprehensive literature review identified technology-dominated CTI discourse as the prevailing paradigm, providing independent confirmation from systematic analysis of published research. This convergent

² APRA – Australian Prudential Regulation Authority ([link](#))

³ CPS 234 – Information Security (Cross-Industry Prudential Standards)

⁴ FS-ISAC – Financial Services-Information Sharing Analysis Center ([link](#))

⁵ Sarbanes-Oxley Act (SOX) of 2002

evidence from action research, industry survey, and literature synthesis suggests FINORG's structural configuration produces systematic rather than idiosyncratic outcomes; organizations facing similar structural conditions appear to generate similar dysfunctional patterns.

FINORG constitutes a "least likely" case for observing intelligence dysfunction (Flyvbjerg, 2006). The organization's resources, expertise, and commitment to CTI substantially exceed most financial institutions: a five-year mature program with dedicated staff, executive-level security leadership, sophisticated security operations, and substantial technology investment. If systematic dysfunctions persist under these favorable conditions, they are likely amplified in organizations with fewer capabilities. This critical case logic establishes a conservative baseline, ergo, what we observe represents a best-case scenario for many organizations. The inverted intelligence model emerging despite favorable conditions implies the problem is structural and inevitable rather than resource-dependent. Organizations with less mature programs, smaller teams, or lower executive attention face these dysfunctions plus additional challenges from capability gaps.

We selected FINORG for several compelling reasons: (1) maturity, as the CTI function has been established for over five years and has reached operational stability; (2) perceived risk exposure, FINORG faces persistent, sophisticated threats from criminal and state-based actors; (3) resourcing, FINORG has invested significantly in CTI capabilities, with dedicated staff, technologies, and processes; and (4) access, the organization provided access to diverse stakeholders across leadership, business, and technical domains, including direct engagement with CTI practitioners. Further, on point (4), the organization possesses a dedicated CTI team with specialized roles and responsibilities, substantial budget for threat intelligence tools and external services, mature sophisticated processes predicated on industry regulator requirements and global technology industry standards, executive support for intelligence-led security initiatives, and integration between CTI and broader security operations.

3.3. Theoretical underpinnings and the stakeholder activity process framework

We seek to develop process theory to *understand* and *explain* the complex phenomenon of CTI practice in organizations. To build new process theory we needed to start with a kernel theory as the basis or reference from which we could draw constructs. Whilst CTI researchers have proposed various cycle adaptations emphasizing technology (Tounsi & Rais (2018)) or practitioner tradecraft (Chismon & Ruks (2015)) (see Section 2.2 for comparison), we chose the foundational four-phase military intelligence cycle formalized across NATO nations as described in Section 1 as it: (1) frames intelligence as a decision-support process which corresponds to our objectives of understanding organizational decision-support; (2) contains arguably the simplest conceptualization of intelligence centered on four broad constructs (direction, collecting, processing, dissemination) making it practically useful for process theory development; (3) is a long-standing and mature process model withstanding the rigors of 70+ years of practice; and (4) is applicable to diverse contexts and situations such as an organizational socio-technical environment.

Our justification for developing *process theory* is: (1) the phenomenon studied, CTI as a decision-support function, is a process - i.e., a series of events occurring within an organizational context; and (2) the process is complex and multi-faceted, so it is best studied through the interpretations of stakeholders. Further, given we aim to improve CTI practice, we argue (see Shaw and Jarvenpaa (1997)): (1) process models provide concrete *practice* insights rather than construct correlations, and (2) process models are easier to visualize and implement than other types of models (e.g., variance) as practitioners can change activities in the organisation rather than independent variables.

We propose to construct a particular kind of process model, i.e., a stakeholder-activity process model, to explain the practice of CTI in

organizational contexts (see the various process models in Alsubaie et al. (2025)). The model is intended to be used as a *diagnostic tool* to develop and/or evaluate the CTI practices in large and well-resourced organisations. As our study is tightly scoped to CTI in organizational practice, we only model the four states of Direction, Collection, Processing, and Dissemination.

To construct the model, we determined how the firm's CTI *capability* is implemented through processes that vertically and horizontally integrate stakeholders across the broad enterprise. Our stakeholder-activity process model is modelled as a two-dimensional artifact (see Table 1). The first dimension encompasses the intelligence cycle phases (Direction, Collection, Processing, and Dissemination) as established in military doctrine. The second dimension identifies the three key stakeholder groups involved in CTI operations: (1) The Leadership Group: C-suite executives and senior security management responsible for strategic direction, (2) the Business Group: Risk management, operations, and business unit security teams who consume intelligence, and (3) the Technology Group: CTI analysts, security operations, and technical teams who produce intelligence.

3.3.1. Acknowledging doctrinal variations and feedback mechanisms

The intelligence cycle is conceptualized differently across military doctrines, though these variations reflect emphasis rather than fundamental disagreement. The Australian Army (Army, 2018), UK Ministry of Defence (MOD, 2023), and NATO employ a **four**-phase model (Direction, Collection, Processing, Dissemination), whilst the US Department of Defense employs a **five**-phase model (JCS, 2013, pp. 1–6 (Figure I–3.)). The UK doctrine explicitly notes these differences as “merely a different emphasis in describing the same activities”(MOD, 2023, p. 39), whilst the US Senate Select Committee on Intelligence (1996) (Lowenthal, 1996) proposed a modified functional flow that similarly encompasses these activities whilst emphasizing different management functions.

Critically, regardless of phase structure, military doctrine emphasizes that the intelligence cycle is iterative and dependent on continuous feedback mechanisms. UK doctrine explicitly states that “Intelligence dissemination should include provision for feedback and dialogue from the decision-maker and users, between peers and from analyst to collector” (MOD, 2023). The Australian Army doctrine similarly emphasizes that intelligence “requires constant review and updating if it is to remain current and relevant to the commander’s needs” (Army, 2018), representing this through “continuous communication and review” connecting all cycle phases. This feedback, alternatively termed ‘evaluation’ or ‘requirements refinement’ in different doctrines, is represented in doctrinal diagrams as surrounding or connecting all phases rather than as a discrete sequential step (see UK MoD Figure 3.1; Australian Army LWD 2–0 Figure 2–1; Chismon and Ruks, 2015 Figure 4).

Historical analysis reveals that feedback has been fundamental to intelligence doctrine since its earliest formal articulations. The 1948 US War Department publication “Intelligence is for Commanders”

Table 1
Stakeholder-activity process model across intelligence cycle phases.

Intelligence Cycle Phase	Stakeholders		
	Leadership Group	Business Group	Technology Group
Direction			
Collection			
Processing			
Dissemination			

Note: This model represents the four core intelligence cycle phases as defined in Australian Army doctrine (Army, 2018) and UK and NATO doctrine (MOD, 2023). Feedback mechanisms are examined separately in our analysis (Section 5.2.2).

established the cyclical nature of intelligence, emphasizing that “the cycle must be completed by getting the intelligence back to those who need it” and that “evaluation of results feeds back into new requirements” (Glass, 1948, pp. 8–11). This foundational principle has remained consistent across seven decades of doctrinal evolution, though its explicit representation varies across contemporary allied publications.

For our stakeholder-activity process model, we adopt the four-phase Australian/UK/NATO structure for three methodologically justified reasons. First, it aligns with our organizational context—FINORG operates within the Australian regulatory and intelligence-sharing environment, where Australian Defence doctrine provides the most relevant template. Second, the simpler four-phase structure provides clearer analytical boundaries for examining stakeholder activities in civilian contexts where organizational roles rarely map to the more granular military intelligence specializations. Third, the distinction between ‘processing’ and ‘analysis’ as separate phases is less meaningful in civilian CTI contexts, where these functions are rarely separated organizationally and are typically performed by the same analysts using integrated technological platforms.

Crucially, feedback is not omitted from our model but rather examined as a cross-cutting mechanism that should connect all phases—a theoretical positioning that, as we demonstrate empirically, reveals systematic feedback breakdowns at FINORG. By analytically separating feedback across the four core phases, we can distinguish between the **prescribed process** (what military doctrine mandates) and the **actual practice** (what civilian organizations actually implement). This separation enables us to demonstrate that feedback loops, though theoretically fundamental and explicitly prescribed in military doctrine, fail to materialize or function effectively when intelligence doctrine is translated to civilian organizational contexts. The absence of functional feedback mechanisms, we argue, is not merely a process deficiency but a structural characteristic that drives the bottom-up inversion and formal-informal divide we observe.

3.4. Data collection and analysis

Our data collection employed multiple methods to achieve triangulation and depth. Semi-structured interviews were conducted with all stakeholder groups and organizational levels (Table 2), lasting 45 to 90 min each. Topics included: their understanding of CTI’s purpose and value within FINORG; specific activities within the CTI program; interactions with other stakeholder groups; challenges and workarounds in daily practice; examples of intelligence processes used in decision-making; and perspectives on CTI effectiveness and opportunities for improvement. The semi-structured format enabled participants to describe their experiences in their own terms while ensuring comprehensive coverage of key topics.

We analyzed the data using a systematic, iterative approach based on qualitative research methods (Gioia et al., 2013; Miles, 1994; Neuman,

2014). Adopting three types of coding as enunciated by (Strauss, 1990), and described by Neuman (2014), our analysis followed the widely accepted coding process of open, axial, and selective coding. The analysis was iterative, with initial findings informing subsequent data collection and analysis. Where clarification was needed, we conducted follow-up interviews with specific participants.⁶

4. Findings

Our primary finding is that FINORG generates requirements within technology operations silos, pushing intelligence outward and upward to operational, tactical, and strategic levels. While Business and Leadership groups are conducting parallel collection activities, and consumers in general across the enterprise are re-interpreting and re-analyzing intelligence from the Technology Group to make sense of the content and apply it to their respective domains. We describe FINORG’s CTI practices phase-by-phase, examining how each stakeholder group engages with CTI practice through Direction, Collection, Processing, and Dissemination.

4.1. The Direction phase

Phase Description. The Direction phase represents the foundational step of the intelligence cycle, establishing what information is needed, why it is needed, and how it is likely to be sourced. This phase involves setting precise requirements, establishing priorities, and initiating planning activities to guide subsequent collection efforts. This phase ensures intelligence operations remain strategically focused and decision-relevant.

Summary Finding. Direction emerges primarily from the Technology Group with only general organizational mission guidance provided by the Leadership Group and specific and ad hoc requests from the Business Group.

4.1.1. The leadership group

The Leadership Group’s participation in Direction proved virtually non-existent (with the exception of the occasional query triggered by media reporting), creating a vacuum that forces other groups to compensate. The CISO remarked, “Board members usually are on several boards at any one time; they’d seen it [threat information] discussed in other companies, and so, they were demanding to know, [are we vulnerable], and why weren’t we doing more here [to counter that threat to us]?”.

Executive leadership expects intelligence to materialize without their input. The Business Operations Executive captured the organizational struggle with requirements elicitation from the program’s inception: “When we went hard at setting up a structured CTI program about five years ago, we did start off by trying to go and elicit requirements, but we found that the organization wasn’t very good at articulating what their intel requirements were. Some of that is culture shock”.

The Senior Strategy Executive confirmed their limited involvement beyond high-level strategy: “So, I’m responsible for the security strategy for the organization. We run a threat intel-informed investment strategy. So, every quarter my team refreshes our strategy... with the Cyber-Threat Intelligence team and understanding their updated assessment of the current threat environment”.

While this quarterly engagement exists, it fails to translate into actionable intelligence requirements that guide daily operations as is evidenced by the Cyber Defense Manager who points out that the CTI team sets up its own workflows while operating against a general mission. The CTI Team does not get specific and actionable requirements arising from strategy meetings: “We review with [the Senior Strategy Executive] what are those requirements from a business level perspective at the

Table 2
FINORG participant profiles.

Stakeholder Group	Role Description	Industry Experience (years)	CTI Experience (years)
Leadership Group	CISO	20	10
Business Group	Senior Strategy Executive	20	12
	Business Operations Executive	17	15
	Strategy & Research Executive	8	1
	Senior Risk Manager	20	3
	Cyber Defense Manager	15	5
Technology Group	Senior CTI Manager	20	12

⁶ All quotations associated with FINORG roles reflect data arising from participant interview transcripts.

very top? So, they're fairly infrequent. They often, they don't change, to be honest with you. The tasking requirements in terms of then as it flows down to the collection requirements, there's not a ticket generated for them. They have a mission to execute against, which is to collect information that is relevant to, which comes from the strategy piece. And then they have on a day-to-day basis automated workflows to gather intelligence, bring that in."

4.1.2. The business group

The Business Group attempts to bridge the gap between absent leadership direction and operational intelligence needs, but struggles with both obtaining requirements from above and translating them for technical teams. The Strategy & Research Executive points to the disconnect with the operational layer, "CTI are not necessarily in those [strategic] decision forums. If it's about security tools, however, they provide the input for it", invariably resulting in a translation burden that falls upon the Business Executive and creates an indirect feedback loop with the security team: "So when I said this is the group view of the key threat scenarios we care about, we have a team of Business Information Security Advisors who will work with those respective business units... they need to effectively have a translation of that to understand basically the cyber risk profile for their division. And so, they'll use that as input/output. And then certainly if there's any feedback, I would expect to hear that from our Business Information Security Advisors". The gap is also reinforced by the observation: "They [the CTI Team] are primarily focused on the threat angle and specifically threat actors. So even in the context of the threat model, that doesn't cover all of the threat scenarios", suggesting that the Business Group's intelligence needs are not facilitated by CTI Team reporting, and consequently, deference is likely afforded towards other Business Group sources.

When the Business Group does attempt to provide direction, it often comes in the form of ad-hoc requests rather than systematic requirements. As the Strategy & Research Executive explained: "If we've identified what we think might be a new threat scenario, I might share that with them [the CTI team] and say, actually, can we keep an eye out on this particular topic and see if there's any signs of escalation?".

4.1.3. The technology group

The ad-hoc requests from the Business Group do not collectively form coherent Direction. As a result, the Technology Group has been forced to become self-directing, generating their own requirements based on technical capabilities and analyst assumptions. The Senior CTI Manager's candid admission reveals the fundamental problem: "Yeah we don't get the direction". This absence of direction forces the team into a problematic position: "...our requirements at the moment are purely just derived from, I think, me and the rest of the team thinking about what sort of information we actually have and what sort of information people should be caring about".

The challenge extends beyond simply lacking direction to not knowing how to elicit requirements effectively. The Senior CTI Manager explained: "So, from the requirements part, it's actually understanding what our customers want. But the hard thing is that I think the challenge that we have is that we don't know how to actually ask the right questions to tell them what they want as well, because we know that the customer never knows what they want". This creates a circular problem where analysts must guess at requirements because stakeholders cannot articulate them, and stakeholders cannot articulate requirements because they don't understand intelligence capabilities.

The self-generation of requirements based on analyst experience rather than organizational strategy was further confirmed: "We [the CTI team] had no understanding [of intelligence requirements], and so I had to make it up based on my experience... So that's going to be one of our requirements then in terms of actually having that experience and talking to people about gathering requirements. We haven't done that yet".

4.2. The Collection phase

Phase Description. The Collection phase involves the active gathering of information to meet intelligence requirements established in the Direction phase. In traditional doctrine, this phase focuses resources on acquiring specific information needed to answer intelligence questions, with collection managers prioritizing efforts based on strategic importance, timeliness, capability, and resource constraints.

Summary Finding. Without clear requirements from the Direction phase, Collection at FINORG is fragmented into three parallel collection streams. The Leadership Group reaches out to informal peer networks, bypassing its own threat intel capability, the Business Group has set up its own collection capability, which is likely redundant to the far more sophisticated capability possessed by the Technology Group that sources intel primarily from open source and commercial subscriptions to specialized threat intelligence feeds but also through trawling internet forums such as the Dark Web.

4.2.1. The leadership group

Unable or unwilling to provide formal direction, the Leadership Group collects its own intelligence, relying heavily on informal relationships and networks. The CISO described this approach: "We might hear something or be told something, or I might get a phone call that says, off the record, I just want to give you a heads up". The Leadership Group uses industry peers as collection sources: "We've got friends and colleagues monitoring and seeing Twitter feeds and we'll get a Signal chat which we're all on and we'll go, hey, just seen this". More often, it's a response occasioned by open-source media: "I'll quickly get asked... do we know who it is? Do we know who did it? What's their MO? Do we understand their capability and how might that impact us? That's a very quick and rapid set of discussions that will happen anytime something's been disclosed in the media and we'll work pretty hard to figure out who is it, what do we know and therefore, what can we ascribe about their capabilities and likely intent and what's likely to happen next".

4.2.2. The business group

The Business Group has its own parallel collection mechanisms to address specific business needs not met by the CTI element within the Technology Group. The Business Operations Executive observed, "Then you'll have stuff that sort of comes in I like to think of it as sideways. And that's more of your collaboration pieces, where it might be participation in your trusted circles or it might be your quarterly InterBank, or what have you, where you're effectively getting information from other peers who are calling out trends or threats or interesting things, and you start doing some of your own digging". An impromptu request can also arise: "We do also get a lot of top-down stuff where our CISO, has seen a thing, or one of the group executives or the CEO has read an article or has had a question about a thing and they pop a request [to] facilitate a response.". The Strategy and Research Executive noted, "CTI has a very informal process of going out there, collecting whatever they think is important, input-wise, and then sending it off to the people that they think need to know.", reflecting the disparity between what the Business Group needs and the communication difficulties that can result in an intelligence collection activity that ultimately doesn't meet their intelligence requirements.

The Senior Strategy Manager confirmed their reliance on external intelligence services: "they [the CTI team] have a set of information sources that's everything from private information sharing communities of various kinds, both kinds of email-based as well as increasingly TIP to TIP links through STIX and TAXII. Plus, obviously, lots of open-source information, lots of open-source reporting about what's going on in cyber". The Senior Strategy Manager also noted a compounding issue in the sheer volume of data available and its inherent complexity, describing a 'curated' outcome, "In the modern world, everybody's already drowning in data and information, and there's a real question of not just every time you bring in a bunch more signal, it comes with ten times [the] noise... so there's this natural tension between how do we get connected up to every possible source of

information set against the fact that we can't possibly consume an infinite amount of information. So, there's a natural tension there that CTI team's kind of need to manage and really build a really deliberate practice of curating on an ongoing basis".

4.2.3. The technology group

The Technology Group has an impressive collection infrastructure ranging from open-source, conventional paid subscription services as well as premium and bespoke collection services. The Cyber Defense Manager detailed their collection sources: "We've got a number of different collection sources. So, we've got a lot of paid collection sources. So, we've got things like Mandiant, BAE, Flashpoint, and the like... but for each one of those, we've actually requested an additional service, which has somebody within their organization doing prioritized collection based on our requirements".

The Senior CTI Manager described the reality of unfocused collection: "At the moment, we've got a standard suite of forums that we go and collect information from, both on the open Internet and also, then some dark web forums". When asked what drives collection priorities, the technology-driven rather than requirements-driven nature becomes clear: "So they've gone through an Intelligence platform, they've got a bunch of processes that they run... The second big kind of big block of tech is [name removed], which is a threat intelligence platform".

The Business Operations Executive observed, "You've got your bottom-up. I like to think about it as where the threat intel analysts are doing their own discovery work and each intel analyst has people have preferences. They'll have their own sources, their own sites. They like to follow everything from a bunch of Twitter feeds through to articles published by threat intel providers, that sort of thing. But I think of that as bottom-up because that's really the intel team gathering activity themselves."

The CISO describes the variety of formal and informal channels used by the CTI Team: "They [the CTI team] work with peer network groups like CISOlens and the Interbank Community, the I-RAP Threat Intel community to share operational intelligence, attack intelligence details about the latest techniques from different groups and operators and then they ascribe labels and names and definitions to what those groups are".

4.3. The Processing phase

Phase Description. The Processing phase transforms collected information into intelligence through analysis, evaluation, and interpretation. This critical phase requires rigorous analytical techniques, structured methodologies, and trained analysts who can distinguish signals from noise and develop insights that support decision-making.

Summary Findings. The Technology Group's processing centers focus on aggregating and curating information using templates. The Leadership group plays no direct role in Processing, and the Business Group focuses on contextualization for its own consumption.

4.3.1. The leadership group

The Leadership Group has no direct involvement in Processing, viewing the transformation of information into intelligence as a technical function rather than a strategic capability requiring their input and understanding.

4.3.2. The business group

The Business Group attempts to process intelligence for their specific context but struggles with both analytical capability and strategic relevance. The Strategy & Research Executive acknowledged the challenge of strategic analysis: "It's really quite difficult. It's a deliberate discipline to take a whole series of those event summaries over, say, a six-month period, and turn it into a view of situational intelligence. Where are we in aggregate? How is the universe moving? And how do we need to adapt to that set of shifts?".

The Business Group uses technology platforms to support processing but primarily for aggregation rather than analysis: "One of the key

outputs from our CTI team is their strategic quality threat assessment... primarily, I use that as an input into the strategy to determine changes in trend direction for the set of threat scenarios we have, as well as identification of whether there are new threat scenarios that we might need to consider".

The difficulty of interpretation is noted by the Strategy and Research Executive: "So when we talk about CTI and threat intelligence, this is largely what's happening in the outside world. That threat environment, how we translate that to our internal organization is the use of that threat model and that's connecting it back to cyber risk as a risk class."

Additionally, the Strategy and Research Executive observed: "So, by exception for things like major events, [such as] Ukraine, Russia, [and] Log4J, they [the CTI team] will generally produce an operational update that will factor into [the] threat model. Similarly, it [is] also factored into the strategic output. So just to your earlier comment around specific IOCs or TTPs or attributing those to threat actors, that gets, I'm going to say grouped up, synthesized and bundled into the quarterly threat assessment is how I see it."

4.3.3. The technology group

The Technology Group conducts extensive processing activities but faces fundamental challenges in analytical methodology and training. When asked about the structured analytical process that transforms information into intelligence, the CTI Manager states: "the most basic thing that I ask people to provide nowadays is when you're writing your tactical report, go through and highlight what you think needs to be included from that original source and tell me how you've planned it and no one gives that to me at all" and "the hard thing is when you're resource tight as well and when people are just trying to just get it done, for example, then people will forget to use it or not want to use it".

When discussing source reliability assessment, the Senior CTI Manager revealed: "So in terms of the source that we actually get this from, we go through the summary... So, this one's 'A', which was reliability, 'A-1'.. So, it's come from Mandiant. But to be honest, I don't think anyone really knows what it means". This admission of using the 'Admiralty Scale' grading system without understanding its purpose suggests FINORG has adopted intelligence framework components without the underlying analytical discipline. A further issue is an apparent willingness to accept the accuracy or truthfulness rating from an external provider without testing (or having the capability to test) its veracity.

The lack of formal analytical training compounds these issues. While the Senior CTI Manager noted that "the actual intel itself... it's not just a report of someone saying something. It's actually, like it does go through that process of someone thinking about it, looking at the trends, and then pulling it all together", indicating that this process relies on individual judgment rather than structured techniques.

This is reinforced by the CTI Manager stating: "...how do we train more intel analysts to be doing actual critical thinking and structured analysis technique, types of intel analysis and not oh, here's an indicator of compromise that's within oh, here's a TTP. How do we train them to be thinkers and not data monkeys?".

4.4. The Dissemination phase

Phase Description. The Dissemination phase delivers relevant, timely, and predictive intelligence to decision-makers in formats that emphasize accuracy and usability. Effective dissemination requires understanding consumer needs, tailoring products to different audiences, and establishing feedback mechanisms to refine intelligence production.

Summary Findings: The Technology Group produces between 5 and 15 daily tactical reports as well as strategic reports using standardized templates and workflows, prioritizing quantity over quality. Consumers across the organization engage in re-interpretation and additional analysis of the content to draw relevant insights, suggesting intelligence products are not having the desired effect by directing organizational decision-making. Evidence from Leadership and Business Groups shows a reliance on informal relationships, particularly peer networks for relevant intelligence, instead of internal products.

4.4.1. The leadership group

A range of intelligence products is consumed by the leadership group. These include a weekly CEO intelligence summary, a Quarterly Board Risk Committee presentation, and Executive Summaries.

The Cyber Defense Manager points to the weekly CEO intelligence summary: On a weekly level, we provide a threat intelligence summary that goes to the CEO. So, The CISO provides a quick email to the CEO every week, and that includes a summary of the most relevant and pertinent bits of information that the CEO would like to read".

The CISO explains that he acts as a translator or interface between the Technology and Leadership Groups, presenting to the Board Risk Committee every quarter: "I present to the board every quarter through the Board Risk Committee a cybersecurity update... I'll talk about the latest threats. So, I'll describe the recent breaches that we've been seeing on other organizations, recent breaches on file transfer appliances. We ascribe them to a threat actor in the report, for example, where we tell them that it was the Klopp Ransomware Group or FIN11".

A third avenue of threat intelligence reaches the Leadership team as Board Paper Executive Summaries as explained by the Strategy and Risk Executive: "the executive summary of that (Quarterly Threat Assessment) section is taken out of the threat assessment and included in board papers".

When intelligence products reach the Leadership Group, they require significant reframing or translation to extract strategic insights. The CISO explicitly acknowledged this translation requirement: "But you can see how we use the kind of CTI or what's happening in the threat space to permeate the way we update the board. But obviously, other than naming the initiatives, we then translate it into plain English for them. So, my team helped produce all of those updates, but then I kind of tell the story, if you like, in plain vernacular, for the Board". The Senior CTI Manager acknowledges the interpretation burden largely falls on the CISO: "The CISO does read our reports and does try to, I think, use our data and interpret our data".

4.4.2. The business group

The Business Group receives intelligence products but frequently finds them misaligned with business needs. The Senior Risk Manager noted the challenge of making intelligence actionable: "Because a lot of the stuff that I find comes from CTI is not directly relevant for FINORG".

The translation barrier reflects another hurdle, as the Strategy and Research Executive observed, "We have a team of Business Information Security Advisors who will work with those respective business units... they need to effectively have a translation of that [CTI information] to understand basically the cyber risk profile for their division.". This insight reveals a key disconnect, reflecting the perspective that intelligence products fail to address business-relevant questions.

To compensate, the Business Group has developed stratified dissemination processes. The Strategy & Research Executive explained: "Here's the decision you're making. Here's the best information we have to support that decision". This translation layer attempts to bridge the gap between technical intelligence and business consumption, but adds complexity without addressing the root cause. A further valuable insight by the Strategy and Research Executive, affirming the stratification relative to the nature and location of the CTI team's deployment: "I absolutely do think it's institutionalized [CTI placement in the SOC]. Whilst they have been in place for years, they are relatively new to the security function. At a tactical level, at an operational level, that can be the part where it's a little bit more ad hoc, where they might get asked for something but then back again at a strategic level that again is fairly industrialized".

The Senior Risk Manager points out that quite a bit of intelligence comes to the Business Group, but that relevance (and therefore actionability) remains the primary issue: "So we receive the threat intelligence updates from the CTI team. We're on their distribution list, so they do about eight a day, four to ten a day, and they've got a daily summary, and then they have a quarterly view as well".

However, as the Business Operations Executive noted, CTI products

have limited impact outside of basic awareness: "I don't think we're at the level of maturity where the broad business really makes use of CTI. To be fair, I think the cybersecurity message more broadly for most of the organization is really around cyber resilient practices... It's really around those tactical behaviors".

4.4.3. The technology group

The Technology Group produces an array of intelligence products, disseminated to consumers at operational, tactical, and strategic levels of the organization (ranging from 5 to 15 tactical reports to weekly/quarterly strategic reports). They can do this by operating a highly standardized production system leveraging formal templates and workflows that prioritize quantity, consistency and report structure over relevance. The Senior CTI Manager described their standard approach: We have a standard template to use for all of our main reports. But all of them it's just basically like you just need the title, you need the summary and then you need the body of the information kind of thing. And the references at the end". Furthermore, "the tactical reports, they go out via email and we will do each tactical analyst should be aiming for like three to five each day. And that's because there's a lot of cyber attacks that we become aware of".

The Senior CTI Manager describes the impact arising from the introduction of a threat intelligence platform to their operating environment, "From the TIP perspective... The first thing is [its] repository to store indicators of compromise and hopefully have that talking to the other machines that need to use it... The second thing that we want to use it for is the raw database of things that we just know about that's happening. Things that don't meet the actual report threshold... And then the last thing is [as] the playbooks tool for the intel team as well".

The workflows drive the dissemination machine as explained by the Senior Strategy Executive: "They use software for workflow. So, all of the kind of the management process around... producing a document, a message, an instruction, an alert and advisory, all of that is coordinated through [software] as a workflow system. And the cyber-threat intelligence product is numbered according to the [software] ticket, which creates really useful traceability for us". Additionally, "the second big kind of big block of tech is [name removed], which is a threat intelligence platform... It replaces some reasonably agricultural implementations of MISP⁷ with a sustainable industrial, well-run tool".

The multi-tiered dissemination is described by the Cyber Defense Manager: "So that's more the tactical 'inform the war fighter' level. So tactical is quite frequent. We'll have anywhere between five to 15 a day of those coming through from the team". "So, actually, at a tactical level, we use a TIP. It's relatively new for us... in terms of IOCs, those types of things, we received all of those things. We deduplicate them all, and then we disseminate them into [software name removed] as well for review. And that's also pushed into enrichment of alerting [on] those types of things."

Strategic-level reports provide summaries to executives for general awareness and are not designed to be actionable: "On a weekly level, we provide a threat intelligence summary that goes to the CEO. So, the CISO provides a quick email to the CEO every week, and that includes a summary of the most relevant and pertinent bits of information that the CEO would like to read".

The same is true for the most strategic report of a range of products, the quarterly report: "And then we start to step up into the monthly and then quarterly reports. These are going to be threat actor reports. Once again, kind of a larger version of the body of events, a summary of the body of events". Yet even these strategic products are acknowledged as: "not really for any particular action, it's just to go through and inform".

The Senior CTI Manager revealed a key disconnect between the threat intelligence products and the consumers: "But I think the challenge that we have at the moment with them [the Executive and Leadership Group]

⁷ MISP – Malware Information Sharing Platform (<https://www.misp-project.org/>)

is that it's like they're trying to reanalyze our data in a way that kind of fits what they think they're looking for rather than trying to, I suppose, purely use our data... And so, at a high level, they're saying to use threat intel, but nobody actually knows that we're using it".

The Technology Group maintains an extensive archive of all the intelligence reports that are disseminated across the organization (note there is no evidence that the reports are actively used in the Processing phase, suggesting the archives may be kept for compliance reasons). The Cyber Defense Manager notes: "And then we keep them all archived within [software name removed] and some other systems as well".

4.5. CTI framework

Table 3 details FINORG's operationalization of CTI across the CTI cycle's phases, categorized by the three key stakeholder groups: Leadership, Business, and Technology. Each activity is mapped to the evidence presented in the Findings section.

5. Discussion

Our study reveals a paradox in modern business environments where organizations invest heavily in CTI to address cyber-threats as a strategic priority while simultaneously creating structural barriers that prevent CTI from delivering strategic value. We first provide a brief overview of the inverted flow of intelligence in FINORG before engaging in a deep discussion of the root causes of the inverted model of intelligence covering (1) the positioning of the CTI function in the organizational structure, (2) the profound gap between Business and IT Groups and how that gap leads to shadow intelligence systems and policy paralysis while technology amplifies said gap and institutionalizes dysfunction in sophisticated-looking processes and outputs, and (3) the prevalence of analytical products that lack the strategic relevance and analytical rigor that create decision value.

5.1. Inverting the intelligence model: A fundamental challenge to the intelligence doctrine

The primary finding of this study is that FINORG's Technology Group self-generates requirements pushing intelligence outwards and upwards to operational, tactical, and strategic levels. Our findings challenge fundamental assumptions in intelligence doctrine about how decision-support systems should operate in organizational contexts. Intelligence theory (whether military (UK MoD (2023); LWD 2-0, (2018) or civilian adaptations (Chismon & Ruks (2015); Tounsi & Rais (2018))) presumes strategic direction cascades downward to guide intelligence production. This presumption embeds assumptions about organizational hierarchy, authority relationships, and the governance mechanisms that enable requirements-driven intelligence. Our case reveals that when intelligence doctrine is translated to civilian organizational contexts lacking these structural prerequisites, systematic inversions emerge that doctrine cannot self-correct. This is effectively an inverted model of intelligence that constitutes a complete reversal of the norms established in the intelligence doctrine.

The inversion begins with the absence of strategic direction in the Direction phase, exemplified by the Senior CTI Manager's admission that Direction is not provided. This void at the foundation of the intelligence cycle creates cascading effects throughout the organization. Without requirements to guide collection, the Technology Group accumulates everything technically feasible rather than what is strategically necessary. Without focused collection supporting defined requirements, processing cannot produce strategic insights. Without strategic analysis, dissemination delivers a technologist's view of cyber-threats, requiring executive translation rather than decision-ready intelligence.

Our selection of FINORG was based on research access rather than a priori knowledge of representativeness. However, ex-post examination reveals independent corroboration of the bottom-up intelligence pattern

Table 3
Stakeholder-activity process model.

Intelligence Phase	Leadership Group	Business Group	Technology Group
Direction	• Provide organizational overview	• Generate ad-hoc intelligence requests • Identify potential threat scenarios • Translate strategic priorities to operational needs	• Generate intelligence requirements based on analyst experience • Derive requirements from available data and knowledge • Direct collection priorities independently • Establish collection workflows without formal tasking • Operate with extensive (paid) collection sources (Mandiant, BAE, Flashpoint, etc.) • Monitor standard forums and dark web sources • Collect through automated intelligence platforms • Manage collection systems
Collection	• Source information through industry & business peer networks • Collect information via informal "off the record" calls • Access information from personal networks independently	• Collect information from multiple sources, including personal, open-source, and commercial services • Access government feeds (ACSC^a, Home Affairs) directly • Subscribe to commercial intelligence services • Gather information through internal business channels	• Collect through extensive (paid) collection sources (Mandiant, BAE, Flashpoint, etc.) • Monitor standard forums and dark web sources • Collect through automated intelligence platforms • Manage collection systems
Processing	• Review threat intelligence for executive consumption • Validate information credibility through peer consultation • Transform intelligence into strategic context	• Analyze information from collection sources • Synthesize collected data using templates • Contextualize technical intelligence against business requirements • Evaluate threat implications for business operations	• Collate information using threat intelligence platforms • Apply inherited source reliability ratings (Admiralty Scale) • Process intelligence through automated workflows • Transform raw data into standardized report formats and templates
Dissemination	• Receive intelligence assessments • Communicate threat intelligence via executive briefings • Reinterpret CTI data and intelligence assessments for strategic decision-making	• Distribute intelligence to business unit stakeholders • Customize intelligence reports for different audiences • Deliver contextualized threat information to decision-makers • Communicate intelligence findings through advisory channels	• Produce multiple tactical reports daily using standard templates • Distribute intelligence across global regions via email and distribution networks • Disseminate weekly CEO summaries and quarterly reports • Broadcast threat alerts internally through automated systems

^a ACSC - Australian Cyber Security Centre (<https://www.cyber.gov.au/>)

across multiple empirical sources. Kotsias et al. (2022) found that CTI requirements are predominantly set by IT operations staff rather than business stakeholders in their action research study of CTI adoption at a commercial organization (referred to as Greenback), documenting the same technology-driven requirements generation we observe at FINORG. Brown & Lee's (2021) SANS industry survey of financial services found technology operations staff predominantly determine CTI collection priorities rather than strategic leadership, suggesting technology-driven intelligence is widespread across the sector. Sauerwein et al. (2021) and Samtani et al. (2020) independently document technology-centric approaches to CTI implementation where operational requirements dominate strategic direction. As Ainslie et al.'s (2023) comprehensive literature review demonstrated, CTI literature is 'dominated by the technology discourse', a pattern our findings reveal persists in organizational practice. Four independent sources; two empirical studies, one industry survey, and one systematic literature review report the same inverted pattern we observed at FINORG. This convergence suggests the dysfunction is structural (emerging from common organizational configurations where CTI resides in Technology Groups) rather than idiosyncratic to FINORG.

5.2. The root causes of the inverted intelligence model

We assert that the inverted model is not an accident or an isolated case but rather an inevitable outcome of particular attributes of the modern business environment. These attributes make it difficult for organizations to realize the value proposition of CTI despite the significant investment in the function.

We present three such structural attributes: (1) the positioning of CTI as an operational function rather than a strategic support function, (2) the disconnect between Business and IT perspectives, thus inhibiting knowledge transfer, and (3) the over-reliance on technology systems over intelligence-trained human analysts.

5.2.1. Positioning CTI as an operational function where it cannot support strategic decisions

As CTI is a sub-function of cybersecurity, it is typically positioned in the Technology Group as one of several operational units. Empirically, At FINORG CTI reports through technology operations, rather than directly to the executive leadership, creating structural separation. The CISO must therefore translate or interpret CTI outputs for the broader leadership team. In this way, the CISO role becomes a structural separator rather than an integrator.

The fact that members of the CTI team are not directly engaged in strategic planning and related conversations with the broader leadership team outside of the CISO suggests that the structural separation is institutionalized in behavior (not just on paper). Without a formal mechanism to engage with the leadership team, CTI focuses on threat hunting and incident response support rather than strategic questions. Their performance is measured in terms of operational outputs (indicators processed, alerts generated) rather than the extent to which they can influence strategic decision-making.

Theoretically, institutional theory (DiMaggio and Powell, 1983) explains this positioning as coercive isomorphism: APRA CPS 234 mandates information security within operational risk frameworks, making technology positioning the compliant choice. In generalized terms, this regulatory constraint applies to all APRA-regulated financial institutions, predicting similar CTI positioning across the sector.

Excluding CTI from leadership decision-making results in an inverted intelligence model, as the CTI function is pushed down to the lowest level of the organization, where it cannot perform a strategic support function. Consequently, the Leadership Group fills the vacuum left behind by intelligence Collection and Direction through informal peer networks. This exclusion in turn leads to a form of cognitive dissonance as leadership continues to invest in CTI capabilities while simultaneously relying on informal channels for strategic intelligence (Argyris

and Schön, 1978). Additionally, this dissonance is a barrier to organizational learning, themes we explore in depth in Section 5.2.2, where we argue that person-dependent and informal networks prevent institutionalization necessary for organizational learning (Crossan et al., 1999).

5.2.2. The broken feedback loop – doctrine's missing link in practice

The military intelligence literature emphasizes feedback mechanisms as fundamental to intelligence cycle effectiveness. Applying the four-phase intelligence cycle as our analytical framework enabled us to identify where feedback mechanisms should connect organizational groups at FINORG. Military doctrine prescribes feedback connecting Direction to Collection, Collection to Processing, and Processing back to Direction, creating iterative refinement (Army, 2018; JCS, 2013; MOD, 2023). However, civilian organizational literature provides limited examination of how such mechanisms function without command authority to mandate compliance. Where military organizations enforce feedback through hierarchical reporting structures, civilian organizations must rely on voluntary collaboration and governance mechanisms, yet research examining whether these alternative structures succeed in practice remains sparse.

At FINORG, these doctrinal feedback mechanisms are systematically absent. Our empirical examination sought to map these prescribed connections against FINORG's actual practice. The systematic absence of these mechanisms emerged as an unexpected finding. At FINORG, feedback mechanisms are systematically absent across all stakeholder interfaces.

We applied the four-phase intelligence cycle as our analytical framework to examine FINORG's CTI operations. Military doctrine assumes feedback mechanisms connect all phases: Direction informs Collection, Collection validates Processing, Processing refines Dissemination, and Dissemination feeds back to Direction. These connections enable the iterative learning that transforms intelligence from unidirectional production into adaptive organizational capability. Our empirical examination sought to map how FINORG's CTI practice aligned with this doctrinal framework. The systematic absence of feedback mechanisms emerged as an unexpected finding rather than a predicted outcome. The following analysis documents where and how these feedback failures occur across FINORG's organizational structure (see Table 4 below).

Feedback Failures. At FINORG, these feedback mechanisms are systematically absent across all stakeholder interfaces. Vertical feedback failures occur at multiple levels: leadership receives intelligence without providing structured feedback on utility, while Business Groups independently re-analyze products rather than requesting improvements. Leadership's reliance on "off-the-record calls" and peer networks reveals that formal CTI products lack value, yet this assessment never reaches producers. Horizontal feedback failures compound the problem: Technology operates automated collection without validation of strategic alignment, while peer networks exchange valuable intelligence informally but bypass formal CTI processes. These parallel failures create one-way flows where CTI products disappear without assessment, correction, or learning.

The Self-Reinforcing Cycle. This broken feedback loop creates a self-reinforcing cycle perpetuating bottom-up inversion. Without feedback, CTI analysts continue producing technically sophisticated but strategically irrelevant intelligence. Leadership, receiving intelligence that fails to inform decisions, increasingly relies on informal networks, further reducing engagement with formal CTI products and eliminating the incentive to provide feedback. Business, receiving no guidance from above and providing no feedback below, defaults to reactive translation rather than proactive requirements definition. Poor intelligence receives no corrective feedback, so it continues to be poor, driving leadership deeper into informal networks, resulting in a self-perpetuating dysfunction.

Organizational Learning Impediment. This feedback vacuum

represents a structural impediment to organizational learning (Crossan et al., 1999), not merely process failure. Where military doctrine creates iterative refinement through mandated feedback enabled by command authority, FINORG's intelligence cycle operates as an open loop: outputs disappear without assessment or correction. Military contexts enforce feedback through command structures: commanders direct intelligence adjustments, theatre-level intelligence mandates reporting changes, and post-operation assessments evaluate effectiveness. FINORG, lacking hierarchical command relationships between groups, must rely on voluntary collaboration, a condition that our data reveal does not exist.

5.2.3. Business-Technology knowledge gap as a driver for the inverted intelligence model

The Core Problem. The Business-Technology knowledge gap drives stakeholders to informal networks that become primary intelligence channels. This structural divide, well-documented in organizations (Guillemette, 2012; Manfreda, 2018; Peppard, 1999); manifests in CTI through mutual incomprehension: Business cannot articulate requirements in technical terms, whilst Technology cannot translate technical intelligence into strategic context. Business views success through market performance and customer experience; whilst Technology focuses on platform continuity and service delivery. Unable to bridge this gap, Business defaults to ad hoc requests whilst Technology self-directs without strategic guidance.

Theoretical Framing. This gap exemplifies a “semantic boundary” (Carlile, 2004) where distinct professional domains possess specialized knowledge resisting simple transfer. Unlike syntactic boundaries addressable through common lexicons, semantic boundaries require transformation: knowledge must be contextualized, not translated. Organizational learning theory (Crossan et al., 1999) suggests such gaps should resolve through iterative feed-forward and feedback processes institutionalizing shared understanding. Our findings reveal these learning mechanisms systematically fail in CTI contexts, explaining the gap's persistence.

Shadow Intelligence and Consequences. This gap produces shadow intelligence systems. Leadership's reliance on informal channels (off-the-record calls and industry peer networks) becomes essential when formal processes fail to deliver strategic intelligence. These shadow networks operate outside organizational governance, creating three critical problems. First, they depend on individual relationships rather than institutional capabilities, making intelligence dependent on individuals. Second, they bypass formal validation processes, potentially introducing unreliable intelligence. Third, their success prevents recognition of formal process failures: when executives receive valuable

intelligence through peer networks, they perceive intelligence as working without recognizing that formal CTI processes contribute little, preventing the organizational learning necessary to address structural problems.

Stratification Burden: The gap drives compensatory stratification: multiple layers translate and repackage intelligence for different audiences. This creates organizational burden through duplicated effort without analytical value, introduces misinterpretation as context erodes through successive simplifications, and inhibits organizational learning by breaking feedback loops. Multiple intermediary layers prevent producers understanding consumer needs and consumers effectively communicating requirements. This communication failure perpetuates bottom-up inversion by preventing the iterative refinement that could align intelligence production with strategic needs.

Policy Paralysis. The formal-informal divide creates policy development paralysis: successful informal practices cannot be formalized without acknowledging formal process failures. Practitioners cannot document workarounds; leadership cannot recognize informal successes without admitting structural dysfunction. Organizational learning remains un-institutionalized, successes remain person-dependent, and failures repeat because root causes cannot be addressed within formal governance structures.

5.2.4. The role of technological systems in institutionalizing the inverted model

The sophisticated technical capabilities at FINORG (threat intelligence platforms, automated collection systems, and analytical tools) do not bridge the Business-IT gap but instead institutionalize dysfunction and make informal workarounds necessary.

Automation in CTI: Organizations invest heavily in CTI technologies, believing that automation will transform their intelligence capabilities and reduce the need for informal processes. At FINORG, the Technology Group operates extensive automated systems for collection and processing, with the Senior CTI and the Cyber Defense Manager each describing how central the TIP is to documenting and formalizing intelligence-related processes. Yet this automation, rather than enabling strategic intelligence, locks in tactical patterns and makes both the inversion and informal workarounds harder to recognize or correct. Automated systems formalize the bottom-up approach by encoding analyst assumptions about requirements into algorithms and collection rules. Without strategic direction, these systems collect based on technical capability rather than strategic need. The formalization through technology makes these patterns appear legitimate and difficult to question; after all, the system is operating as designed.

Velocity Trap: Modern CTI platforms promise real-time threat intelligence, creating organizational expectations for speed that further undermines both strategic analysis and practitioner initiative. The pressure for rapid intelligence delivery, amplified by automated systems capable of processing millions of indicators daily, creates a ‘velocity trap’ as it leaves no time for the deliberate analysis that distinguishes intelligence from information. More perniciously, it suppresses the analytical initiative that might question whether speed serves strategic purposes.

The Strategy & Research Executive describes how challenging it is to develop a view of situational intelligence from event summaries collected over a six-month period. This deliberate discipline cannot be automated and requires initiative to pursue despite organizational pressure for immediate tactical responses. Yet formal metrics reward speed over insight, causing practitioners to suppress strategic initiative in favor of meeting velocity expectations.

Technology enables FINORG to report CTI activity with unprecedented precision, numbers of indicators processed, alerts generated, reports produced, threats detected. These metrics become formalized in reporting cycles. Yet they simply report activity rather than measure value, creating an illusion of intelligence effectiveness that obscures both the absence of strategic value and the informal practices that

Table 4
Feedback mechanism failures at FINORG.

Feedback Type	Doctrine Prescription	FINORG Observed Dysfunction
Decision-maker to Analyst	Leadership evaluates intelligence relevance, communicates refinement needs	- No structured feedback mechanism - Leadership uses informal networks without informing the CTI team
Consumer to Producer	Consumers identify gaps, request adjusted priorities	Business independently re-analyze rather than requesting improvements from CTI
Analyst to Collector	Analysts validate sources, adjust collection priorities	Automated collection operates without validation feedback on strategic alignment
Peer-to-Peer (Horizontal)	Collaborative feedback between units improves collective intelligence	Occurs informally through shadow networks, bypasses formal CTI processes

Note: Military doctrine prescribes continuous feedback enabling iterative refinement. FINORG exhibits systematic feedback absence across all stakeholder interfaces, preventing the organizational learning necessary for intelligence-driven decision-making.

protect the organization. This ‘measurement illusion’ stifles the development of CTI-driven strategic cybersecurity behavior. When formal metrics show impressive activity levels, there appears to be no problem to solve and no actions to take to improve cybersecurity. The informal networks, personal relationships, and workarounds that deliver value remain invisible to governance structures focused on quantitative metrics. This invisibility prevents informal successes from being recognized, studied, and institutionalized into improved formal practices.

5.3. Curated intelligence or true intelligence? The false confidence problem

The combination of bottom-up inversion, formal-informal divide, and technological amplification produces what we call ‘curated intelligence’, namely, analytical products that meet formal requirements and pass through sophisticated technical systems but lack the strategic relevance and analytical rigor that create decision value.

Curated Intelligence. Curated intelligence is driven by sophisticated TIPs, template-driven processing, time pressure and no formal training in intelligence tradecraft among CTI analysts. The existence of ‘intelligence-like’ analytical products creates a capability illusion, leading to false confidence in the intelligence function. This allows for strategic blindness and erosion of long-term resilience rooted in poor engagement with high-value intelligence that frequently comes from informal/human relationships (e.g., from professional networks). The problems associated with managing the sheer volume of collected information are recognized by the Senior Strategy Manager, who emphasizes the need to “*build a deliberate daily practice*”.

Curated intelligence emerges at the intersection of formal compliance and informal reality. Formally, FINORG produces extensive intelligence that follows established templates, uses formal terminology, and moves through approved processes. Informally, practitioners know these products lack strategic relevance, which drives the parallel informal networks and ‘reanalysis behaviors’. The production of curated intelligence becomes self-reinforcing. Analysts, lacking strategic direction, produce what they can defend formally, such as technical indicators, threat actor profiles, and vulnerability assessments. These products satisfy the formal requirements for intelligence production but fail to address strategic questions. Leadership receiving formally correct but strategically irrelevant intelligence, turns to informal networks for actual insights, further reducing incentive to provide formal strategic direction, significance, or business relevance without human direction that the bottom-up inversion fails to provide.

Strategic Blindness. The formal-informal divide creates a particular ‘strategic blindness’. FINORG has become exceptionally capable at detecting and documenting threats that fit formal frameworks. In fact, FINORG can demonstrate mature CTI capabilities through formal assessments, already possessing the processes, technologies, and outputs that frameworks require. However, FINORG is likely blind to strategic threats that emerge through informal channels or require initiative to investigate. There is a potential danger that once sophisticated adversaries understand this blindness, they may deliberately operate in the gaps between formal detection capabilities and informal network coverage.

Erosion of Long-Term Resilience. The formal-informal divide systematically erodes organizational resilience by preventing institutional learning and capability development. Where high-value intelligence value flows through informal channels dependent on individual relationships, the organization becomes vulnerable to personnel changes. When formal processes are known to be ineffective but cannot be acknowledged as such, addressing this gap becomes a challenge. This erosion is particularly concerning for long-term security posture. As the threat landscape evolves, organizations need adaptive capacity that comes from institutional learning. The formal-informal divide prevents this learning by creating parallel realities that cannot be reconciled within existing governance structures.

6. Conclusion

Although our findings come from an in-depth study of a single case organization, we argue that the patterns we have observed are likely representative of broader industry reality. This is due to isomorphic pressures that embed bottom-up structures into foundational CTI practice. First, we argue that CTI practices cannot develop in isolation given cybersecurity professionals are highly mobile and take their work practices wherever they are employed. Further, large organizations gravitate towards standardization of roles especially where said roles are designed to attract short-term employment (as is the case with many CTI roles). Second, technology vendors design their platforms for common use cases, making assumptions about roles, use-cases, and workflows. Third, regulatory requirements and industry standards (NIST, ISO etc.) drive convergence among organizational structures and processes. Taken together, these isomorphic forces create a self-reinforcing cycle that institutionalizes the inverted intelligence model through the hiring of standard CTI roles, the adoption of technology with a particular platform design, and the application of regulatory frameworks.

The implications of our findings extend beyond individual organizations to challenge industry-wide assumptions about intelligence-led security. If the bottom-up inversion we document is widespread among organizations that have made strategic investments in cybersecurity, then organizations may face a significant gap between their intelligence expectations and capabilities. Further, that certain attributes such as the positioning of cyber as an operational technology function, the disconnect between Business and IT perspectives leading to the particular power distribution where Business stakeholders are treated as first class citizens as they are revenue generators whereas IT stakeholders are treated as second-class citizens as a cost-center, in effect ensuring that CTI cannot graduate to a strategic function and cannot serve its strategic support purpose. In an era of sophisticated cyber threats from nation-states, organized crime, and advanced persistent threats, such a gap represents a systemic vulnerability with potentially catastrophic consequences.

The following are our contributions to practice and theory.

6.1. Contributions to practice

For practitioners, our findings offer critical insights into why CTI programs struggle to deliver promised value despite substantial investments. Understanding the bottom-up inversion and its consequences enables organizations to recognize and address the multi-dimensional complexity of operating CTI across strategic and operational levels as well as business and technology domains. For researchers, this study opens a new domain of empirical investigation into the gap between intelligence theory and practice, establishing foundations for future research into this critical but understudied phenomenon.

We make several contributions to practice in this study. First, the stakeholder-activity process model (represented in Table 3) is a powerful tool for diagnosing bottom-up operationalization of CTI practice. The model can be used to map intelligence flows arising from key stakeholders, processes and practices, and redesign these to improve strategic coherence and relevance to key stakeholders. Second, our study directs organizations to recognize and legitimize shadow intelligence practices by identifying where the high-value intelligence resides and flows to develop better governance models that can deliver on the promise of the intelligence function (e.g., hybrid structures that acknowledge informal peer networks without bureaucratizing them).

Third, our study directs organizations to consider the technology amplification effect of CTI platforms to better understand the impact on CTI practice, as the insights can be helpful in selecting appropriate CTI platforms and designing workflows in organizations that complement existing governance structures and culture. Fourth, a key contribution of this study is the discovery that significant repackaging of intelligence is occurring among stakeholder groups due to poor relevance and strategic

coherence. Organizations can learn from this insight by examining how the diverse consumers of intelligence engage in reanalysis and reinterpretation and the burden placed on organizational processes. This study can lead to new mechanisms for improved translation of intelligence (see Kotsias et al., 2021 for a clinical study that introduces ‘intelligence handlers’ as intermediaries between Business and Technology Groups to achieve this).

6.2. Contributions to theory

Our findings contribute to three theoretical domains. (1) To intelligence theory, we demonstrate that intelligence doctrine developed within military command structures fails predictably when translated to civilian organizational contexts lacking equivalent authority relationships and feedback-enabling governance mechanisms. (2) To organizational learning theory, we reveal how the absence of feed-forward and feedback processes prevents institutionalization of intelligence practices, creating persistent rather than temporary dysfunctions. (3) To institutional theory, we show how organizations decouple formal intelligence structures from actual decision-making processes; a form of technical decoupling (not merely ceremonial) where sophisticated capabilities exist but remain disconnected from strategic functions they ostensibly serve.

We also make several other contributions to theory. First, we contribute a stakeholder-activity process model that explains how a well-resourced and mature organization operationalizes CTI while uncovering the multi-dimensional intelligence requirements and tensions that emerge when serving both strategic and operational levels as well as business and technology domains simultaneously. This is perhaps first in-depth study of practice of its kind. Whereas previous studies of CTI have focused on technology development and/or information sharing outside of the organizational context, this case study takes a socio-technical view of a single organization to provide a rich view of real-world practice. We detail the role of three key stakeholders – Leadership, Business and Technology Groups explaining how the interact and where the lines of tension and disconnect occur. This approach has enabled us to discover the inverted intelligence model as well as the deep disconnects that explain why this model has become institutionalized in the case organization (and likely the broader industry).

Second, through the identification of sources of information and knowledge and construction of communication pathways and relationships between stakeholders we have developed an information processing network that explains how organizations control information flow to develop threat intelligence. Further, we have contrasted this information processing network with that of military organizations that have a more hierarchical top-down flow. We hope this contribution will attract organizational scientists to further study the contrasting models and further develop this perspective to improve CTI practice in civilian organizations.

Third, we contribute to the discourse of technology adoption in organizations. It is widely known that the adoption of technology has profound impact on the practice of organizations however this study challenges assumptions about the true effectiveness and utility of technology insofar as it relates to effectiveness of threat intelligence.

Fourth, we contribute to process theory by demonstrating how the absence of feedback mechanisms transforms the intelligence cycle from an iterative learning process into a unidirectional production pipeline. Whilst feedback is explicitly prescribed in military intelligence doctrine across allied forces (Army, 2018; JCS, 2013; MOD, 2023) and has been fundamental since intelligence doctrine's earliest formal articulation (Glass, 1948), we show that in civilian contexts lacking clear hierarchical command structures, these feedback loops fail to form organically. This finding extends organizational learning theory (Crossan et al., 1999) by demonstrating that feed-forward and feedback processes—essential for institutionalizing learning—may not develop

naturally when military doctrine is translated to civilian organizational structures without corresponding authority relationships. We reveal that feedback failure is not merely a process deficiency but a structural characteristic emerging from the intersection of bottom-up operations, Business-Technology knowledge gaps, and voluntary rather than mandated collaboration. This theoretical insight challenges assumptions that intelligence doctrine can be successfully adopted in civilian contexts through technical implementation alone, revealing that the social and authority structures enabling feedback in military contexts have no civilian equivalent and require fundamentally different governance mechanisms.

6.3. Limitations

While security and sensitivity issues limited our access to certain aspects of CTI operations, the problems of analyst shortages and the constant high levels of vigilance and activity meant staff were not easily reachable. We could not examine classified intelligence products, observe sensitive operational activities, or analyze actual threat data that might show how formal and informal intelligence differ in content and quality. Although participant descriptions and sanitized examples provided enough insight to identify the bottom-up inversion and formal-informal divide, deeper access over a longer period might uncover additional patterns in how technology systems encode formal dysfunction or how informal networks assess and validate intelligence.

Our framework examines stakeholder activities across intelligence cycle phases, providing a comprehensive view of CTI operationalization. However, this framework may not capture all dimensions relevant to the formal-informal divide and technology's role. Factors such as vendor relationships (which may drive technology adoption regardless of effectiveness), regulatory requirements (which may mandate formal processes known to be ineffective), and organizational politics (which may prevent acknowledgment of formal failures) undoubtedly influence CTI operations but fall outside our analytical boundaries.

6.4. Future work

Our findings, such as bottom-up inversion, establish systematic patterns within large compliance focused, financially regulated institutions. Future research beyond this scope (smaller organizations, less regulated sectors, varying maturity levels) would illuminate how FINORG's patterns adapt under different structural constraints, not whether patterns exist, but how contextual factors moderate their manifestation.

Future research should expand this investigation along two dimensions. First, “breadth expansion” through comparative cases across financial institutions of varying size, regulatory jurisdictions, and CTI maturity levels would illuminate how organizational context moderates the patterns we observe. Second, “longitudinal studies of practice evolution” could capture whether organizations overcome the dysfunctions we document or whether they persist as structural characteristics of civilian CTI implementation. Critically, such expansion would build upon, not validate, the foundational understanding this revelatory case establishes. As (Siggelkow, 2007, p. 20) notes, revelatory cases “can be a perfectly valid source of theory creation,” with subsequent research enriching rather than confirming initial theoretical insights. Our single-case depth provides the necessary foundation; future multi-case breadth would extend theoretical scope.

CRedit authorship contribution statement

Scott Ainslie: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Atif Ahmad:** Writing – review & editing, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Dean Thompson:** Writing – review & editing, Supervision, Conceptualization.

Sean Maynard: Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

There are no conflicts of interest.

Data availability

The authors do not have permission to share data.

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